

Electromagnets

Wind a coil, add an iron core, switch on the current — and you have a magnet you can turn on and off.

Making a temporary magnet

A coil of wire carrying current behaves like a magnet. A **current-carrying coil that behaves as a magnet is called an electromagnet**. Putting an **iron nail (core)** inside the coil makes it much stronger.

Activity 4.2/4.3: when current flows, the coil/nail attracts iron paper clips and deflects compasses; switch off → it loses its magnetism and the clips fall.

An electromagnet has two poles

Like a bar magnet, an electromagnet has a **North** and a **South** pole (Activity 4.4). Reversing the current direction **reverses the poles**.

How to make it stronger

- **More current** (use a battery of more cells) → stronger magnet, bigger deflection, more clips.
- **More turns** in the coil → stronger magnet.
- **An iron core** inside the coil → much stronger magnet.

Lifting electromagnets hang from cranes in scrap yards: switch ON to lift iron/steel, switch OFF to drop it. Their big advantage is they can be turned on and off.

Where students lose marks

Trap 1 — More cells = stronger (not weaker). A single cell gives a weak field; more cells give more current and a stronger electromagnet. Beware "single cell attracts more clips" — that is FALSE.

Trap 2 — The core should be iron (soft), not steel. Iron makes a strong temporary magnet that loses magnetism when current stops — that's what we want.

Trap 3 — Reverse current reverses poles, it does not switch the magnet off.

Without the core: the coil alone still acts as a magnet, but a weaker one — deflection is less than with the iron nail.

Model answers — write them like this

Example A. Give two ways to increase the strength of an electromagnet.

Way 1: Increase the current (use more cells).

Way 2: Increase the number of turns in the coil (or add an iron core).

∴ More current, more turns, or an iron core → stronger electromagnet.

Example B. Why is an electromagnet useful in a scrap-yard crane?

Step 1: It can be switched ON to lift heavy iron/steel.

Step 2: It can be switched OFF to release the load wherever needed.

∴ It can be turned on and off, so it lifts and drops metal on demand.

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